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CatS: A predictive and user-friendly framework based on machine learning models for the screening of heterogeneous catalysts

Gabriel Sardinha de Oliveira [a], Henrique Poltronieri Pacheco [b,*]

[a] Universidade Federal de São Paulo, Instituto de Ciência e Tecnologia, Av. Cesare Monsueto Giulio Lattes, 1201 - Eugênio de Melo, São José dos Campos - SP, 12247-014, Brasil

[b] Universidade Federal do Rio de Janeiro, Programa de Engenharia Química/COPPE/UFRJ, Centro de Tecnologia, Bl. G, Ilha do Fundão, Rio de Janeiro, RJ 21941-598, Brasil

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ABSTRACT

With the machine learning evolution around the world, the application of these technologies has spread to several multidisciplinary areas. In heterogeneous catalysis it was no different. Studies that cover multiple fields have been developed applying these techniques; however, they are not widely employed to experiments due to mutual communication and understanding of problems between these two areas. In this work, a new way of developing, optimizing and presenting convolutional neural networks to the scientific community is proposed in addition to discussions about the problematic use of these tools. With accuracies above 90%, and surpassing previous works, the user will be able to infer results regarding usual chemical reaction systems, such as water-gas shift, selective CO oxidation, and steam reforming, without an exhaustive bibliographic or experimental research to evaluate the best combination of reaction parameters and catalyst features.

1. Introduction

Over the years, several areas have been improved by machine learning (ML) techniques. Regression methods, outlier detectors, or even algorithms for pattern recognition can upgrade task accuracy, assist human decisions, and reduce the amount of time and resources spent [1]. With the application of ML techniques, specifically supervised learning models, it was possible to revolutionize the way of inferring chemical reaction results. Before this connection between ML and heterogeneous catalysis (HC), it was necessary to conduct extensive literature research to optimize any given reaction, searching for and experimentally validating environmental conditions and variables. Another less efficient way of optimizing a reaction is by trial and error, evaluating multiple combinations of variables, which costs resources and time, even with adequate statistical planning of experiments [2,3]. The way that ML models work is similar to this second approach. By trial and error, the model will keep fitting these suggestions according to the previously acquired data (provided by the user or fetched from a database), becoming a powerful and renewable tool, since it is possible to continue to populate the dataset *a posteriori*.

Heterogeneous catalysis literature contains a sizable work that uses ML as support, focused on two procedures: i) Finding which is the best catalyst composition and structure for a certain reaction by studying descriptors [4], crystal faces [5], process variables (i.e., time-on-stream, temperature, preparation methods), among other factors; and ii) understanding and quantifying which property(ies) is(are) relevant for the assessed reaction metrics [6]. This interdisciplinary relation began with a study, dating from 1990, which used artificial neural networks (ANNs) as a physicochemical estimator [7]. Later on, some hallmarks can be noticed in this interaction between the fields of research. Günay and Yildirim [8] applied ANNs and other approaches to quantify the importance of reaction parameters and to predict outliers under untested conditions. Zavyalova et al. [9] and Ahneman et al. [10] focused on developing and validating predictive models for the oxidative coupling of methane (OCM) and C-N cross-coupling, respectively. Regarding studies that used the most varied ML approaches, we reach the current growing scenario of atomic-level studies and reactions databases (Catalysis-Hub.org [11], Materials Project [12], HTEM [13]), reactions with multiples outputs, and a myriad of other possibilities. Nevertheless, some problems often arise in the ML x HC interface:

i Data quality: Each experimental group, due to experimental knowledge, requirements, and literature research, may create a unique

* Corresponding author.
E-mail addresses: gsoliveira@unifesp.br (G.S. de Oliveira), henrique@peq.coppe.ufrj.br (H.P. Pacheco).

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Table 1
Data and sources relation, showing the data granularity.

Authors	Reaction	Data Points	Sources
Odabaşı, Ç., Günay, M. E. & Yıldırım, R.[17]	Water-gas shift	4360	84
Günay, M. E. & Yildirim, R.[8]	Selective CO oxidation	5610	80
Baysal, M., Günay, M. E. & Yıldırım, R.[18]	Steam reforming of methane	5508	81
	Total	**15,478**	**245**

combination of inputs and outputs. This may result in a spaced and, sometimes, incomplete experimental dataset.

ii Data availability: Despite the large amount of experimental work in HC, each study concentrates a small portion of samples. Therefore, it is necessary to conduct intensive research to collect, cluster, and treat these data to make them usable within a ML approach.

iii Knowledge limitation: Even with the implementation of techniques that bring these two areas closer, such as Density Functional Theory (DFT) [14,15] studies and the large use of simulated data [16], the knowledge limitation on both sides can hamper this interaction. On the ML side, there is a difficulty of understanding chemical phenomena in general, and how certain properties affect the experimental results. On the HC side, new computational technologies are constantly launched and present high technical specificity and complex language, which make learning laborious and, consequently, their application in experimental studies scanter.

However, a key problem with the ML x HC interaction is how to make this connection between the experimentalist community and the programming language such that the final user does not need extensive knowledge in both areas. In this work, we propose a way to apply convolutional neural networks (CNNs) in a user-friendly interface (which we named CatS) to predict chemical reaction results and to avoid extensive literature research accompanied by costly experimental validation. Water-gas shift (WGS), selective oxidation of CO (SelOx), and steam reforming reactions were selected to be implemented and optimized. Water-gas shift, which is a reaction to adjust the CO/H_2 ratio in syngas, and SelOx, which is a reaction that assists in the removal of residual CO in pure hydrogen streams, provide the conversion of CO as a result. Steam reforming of methane, chosen to diversify the reaction scope in this work, is one of the most common routes of H_2 production and provides methane conversion as result. For comparison, previous experimental work was used as validation [8,17,18], besides some toolkits to estimate the complexity versus efficiency implementation rate.

2. Methods

2.1. Data selection

Table 1 displays the datasets employed on CatS training. All datasets were pre-treated, removing unwanted variables, such as redundant columns or columns with few occurrences or completely zeroed, and formating the data for use in the network training. Later on, this dataset was divided into 70% of training samples and 30% of validation samples. This technique is seen as the best practice in ML training, since through the division and shuffling of samples it is possible to reduce data

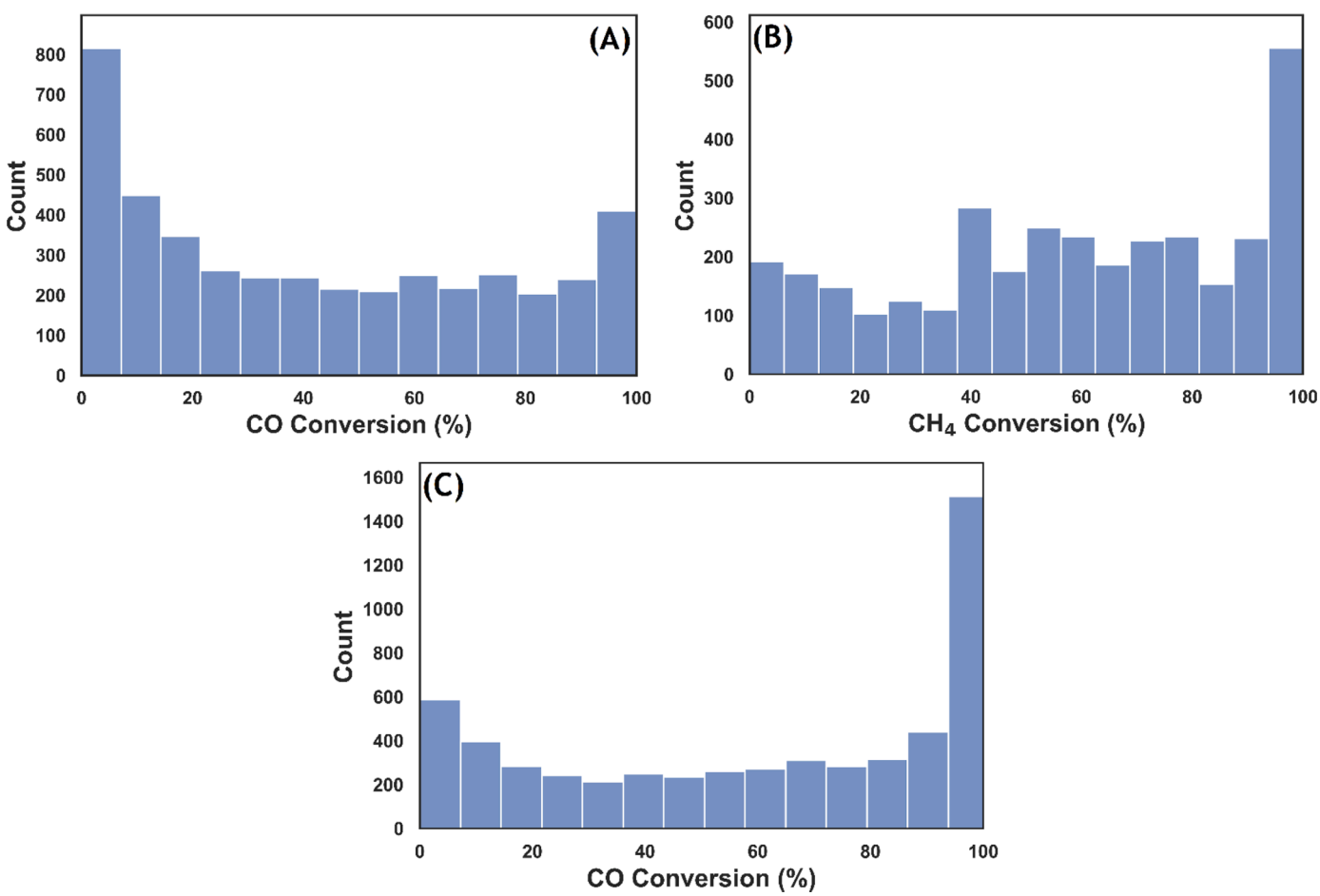

Fig. 1. Data distribution for output values of WGS (A), steam reforming (B) and SelOx (C).

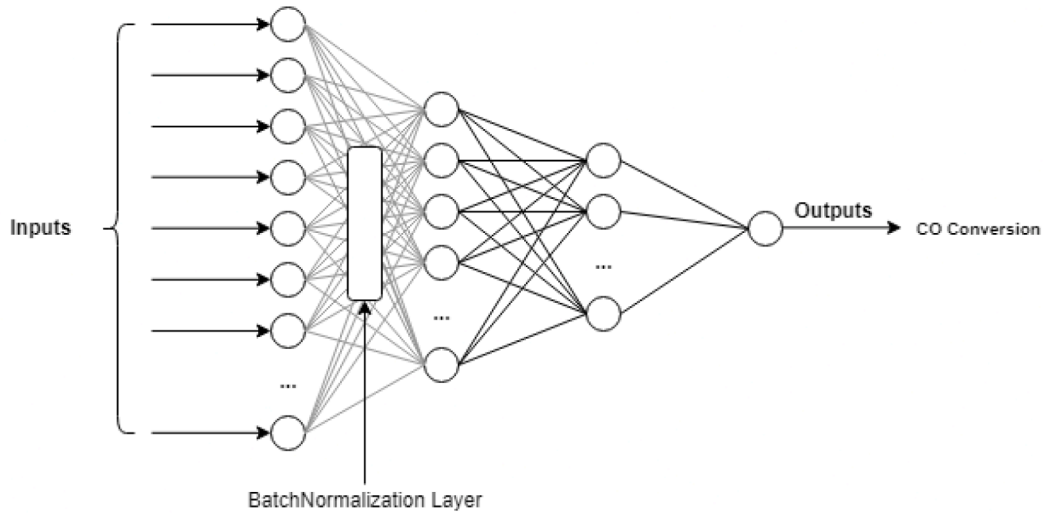

Fig. 2. Neural network architecture for WGS reaction model.

bias and to evaluate the model performance in two different perspectives. For this purpose, a scikit-learn [19] function, responsible for dividing and shuffling the data, was used (train_test_split).

Fig. 1 summarizes the distribution of experimentally reported conversion values used in the training of each reaction model. The input variables, in a HC context, can be divided into two groups: one related to numeric variables (temperature, TOS, calcination time/temperature, etc.) and other to categorical variables (type of reactor, support, catalyst preparation method, etc.). In some circumstances, there may be variables that interfere in the network training (vide infra), so it is important to remove them. Tables S1, S2 and S3 exemplify the reaction variables and outputs used for each model.

In case the user wants to employ experimental data for the creation or improvement of a model, there are some necessary modifications in order to make the dataset usable. The first step is to transform categorical variables into numeric ones, for which a one-hot conversion technique can be applied. The next step is to check the name of each variable and, if necessary, rename it for easier understanding in next occurrences. Finally, fine-tuning the data (e.g., removing unreliable data, rows with many zeros and outliers) can increase the model accuracy. PySpark [20] or pandas [21] can be used as processing tools in this stage that preces the training. All data structuring and treatment in this work was written in Python using PySpark.

The proposed application can use experimental data as training source, either provided by the user or collected in the literature, and in the future it will be able to use simulated data. The user may provide data to complement existing models or suggest the implementation of new models based on data following the pattern described.

2.2. Neural network

We employed an N-layer feedforward neural network to model this entire project, that is, each layer transforms and propagates the information towards the output layer, keeping the same direction, as exemplified in Fig. 2 for the WGS model. We employed the Keras API, using Python as the base of the model architecture, because of its customization level, plurality features (e.g., TensorBoard, TFLite), and integration with Flutter, the framework used to develop the user interface (Section 2.4). Due to the variability of chemical reactions and the multiplicity of environmental variables, each reaction has its specific model with an optimized set of hyperparameters (Section 2.3). Nevertheless, there are two constraints on every model: The neural network (NN) optimizer and the batch normalization layer after the input layer. We applied the Adam [22] algorithm as optimizer, since it is computationally efficient and handles gradient problems. According to Goodfellow et al. [23], batch normalization provides an elegant way of reparameterizing almost any deep network and this reparameterization significantly reduces the problem of coordinating updates across many layers. All the creation and optimization code for each reaction as well as the implemented reaction models for use are available in the CatS GitHub address at: https://github.com/gtbriel/CatS.

In order to evaluate the error in NN learning, a metric used to calculate the magnitude of difference between the prediction and the true value, the mean absolute error (MAE) was employed. Another important metric to support in the learning process is the neural network loss/cost. According to Goodfellow et al. [23], the cost function reduces all the various good and bad aspects of a possibly complex system down to a single number, a scalar value, which allows candidate solutions to be ranked and compared. The employed metric to evaluate the accuracy of the neural networks was the coefficient of determination (R^2), applying the pre-established scikit learn function (r2_score). This measurement is used to explain how much variability of one variable can be caused by its relationship to another related factor. Taking $\widehat{y}_i$ as the predicted value of the i-th sample, y_i as the corresponding true value, and n as the number of validation samples, the estimated R^2 is defined as:

$$R^2(y,\widehat{y}) = 1 - \frac{\sum_{i=1}^{n}(y_i - \widehat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \overline{y})^2}$$

where $\overline{y} = \frac{1}{n}\sum_{i=1}^{n} y_i$ is the arithmetic throughout the y_i.

2.3. Optimization framework

As each reaction has its own model, a generic procedure to create optimized models was developed as a way to design an approach, from CatS' inception, that could easily be extrapolated to new reactions, depending upon the user's demands. This procedure contains a fixed number of trials (100), and each trial generates a temporary model from Optuna [24] hyperparameter suggestions. The hyperparameters suggested by Optuna are learning rate, number of layers, and number of nodes inside each layer. Each temporary model is then trained and validated using the same dataset, and the one with the highest accuracy will be saved. Moreover, each trial can be pruned at the beginning of its training based on the history, that is, if a tested model is considered inefficient, the next generations with comparable results will have their training process interrupted. Valuable information can be extracted from the Optuna's strategies, such as hyperparameter importance for the

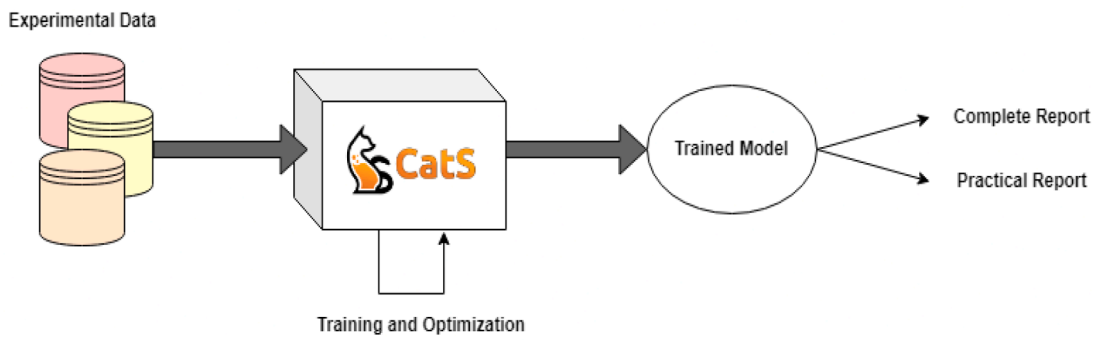

Fig. 3. User's first contact window with the application.

Fig. 4. User's first contact window with the application.

model and the optimization history plot. In order to complement the optimization study, a parameter importance search using ELI5 was added. In short, this library evaluates each network parameter in isolation and captures the accuracy metrics.

2.4. Software interface

As an ongoing effort to make the HC community more intimate with ML techniques, we developed an interface that presents these models in the simplest way, while maintaining academic rigor. To develop the front-end, that is, the section that the user will have contact with, we chose Flutter [25]. It is an open-source framework by Google for building multi-platform and natively-compiled applications from a single codebase, written in Dart [26]. This application does not require an active internet connection and can run on practically any machine, independently of hardware power. Furthermore, all files are compiled in a specific CatS folder for each reaction, allowing for the installation of multiple modules and for user-defined file organization. Fig. 3 outlines CatS' end-to-end operational process, while Fig. 4 illustrates the simplified interface implemented, where the user is asked to select the desired reaction or to search for others. The interface is capable of interpreting any model compiled in TFLite format and that is following the input column formating standards. The user will have two report options at hand: (1) software inference with additional information on the screen; (2) generate a document with the inference supported by all information related to the dataset, creation process and final network parameters and metrics.

Table 2
Overall performance metrics for each reaction.

Reaction	R^2	Loss	MAE
Water gas shift	0.935	0.0068	0.0519
Selective CO oxidation	0.915	0.0111	0.0619
Steam reforming of methane	0.941	0.0053	0.0464

3. Results and discussion

Fig. 1 is already an important source of information, even if it could be assessed at the beginning of a study. It depicts the conversion range obtained by previous experimental reports, which is relevant to understand how the scientific community approaches a given reaction. For instance, Fig. 1A displays a significant density of studies with conversions below ~7%, which may suggest an effort to probe differential conversions, typically employed in researches that aim at studying the kinetics of the first reactions steps. Likewise, Figs. 1B and 1C show multiple studies at the vicinity of 100% conversion, which may raise concern about the thermodynamic constraints of the given systems.

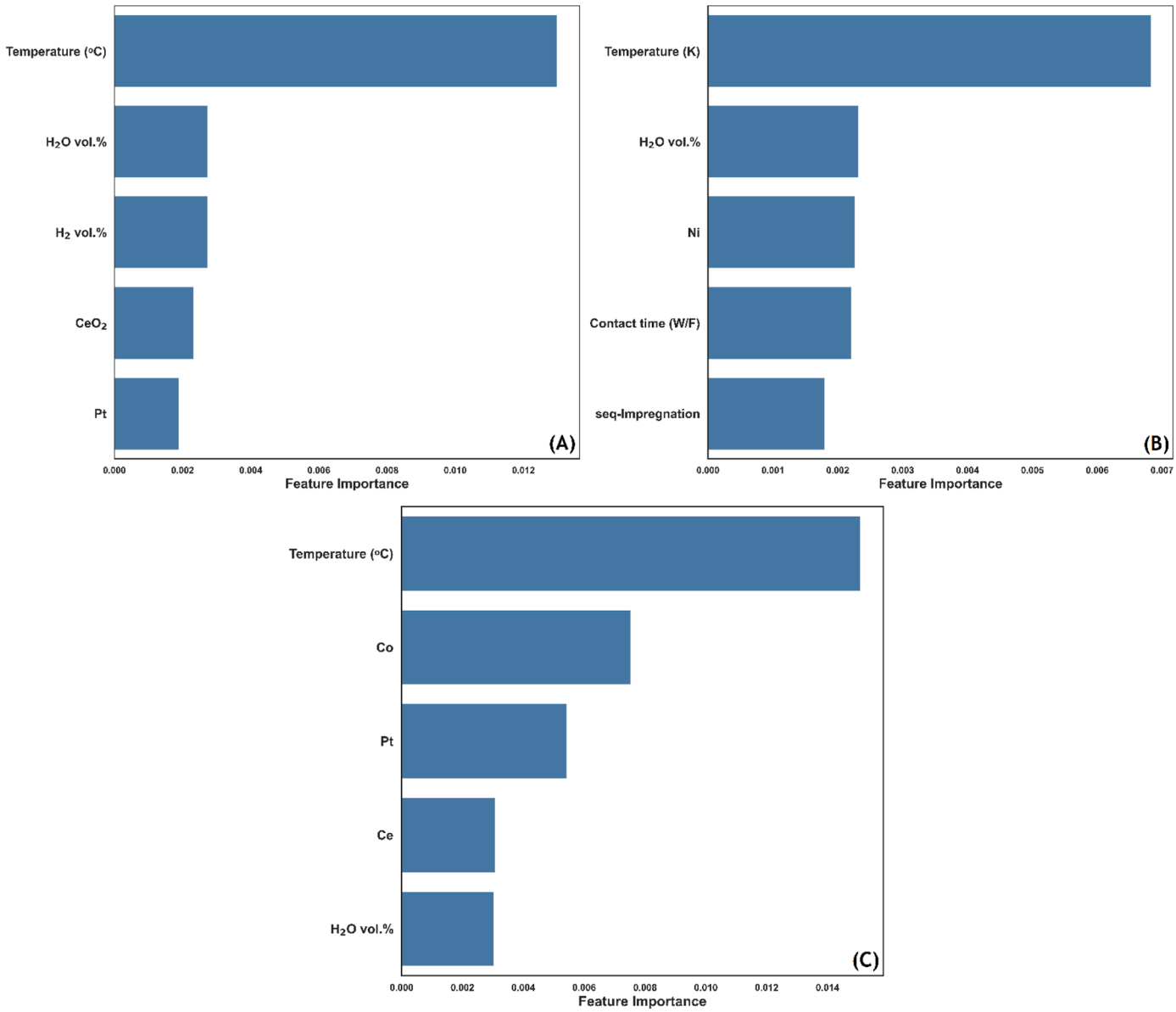

Fig. 5. Most important parameters to WGS (A), steam reforming (B) and SelOx (C).

Table 2 reveals the performance metrics of each implemented model. It shows that all neural networks are optimized and have an accuracy above 90%. These metrics were collected using only the validation samples, that is, 30% of the dataset. They were selected according to consolidated studies [27,28] and based on the demands of the optimization process. Complementing the performance metrics, Fig. 5 shows the five most important variables for the models, and a complete version of this parameter importance plot is shown in the SI (Figures S1-S3), along with plots referring to the optimization process (history and importance of hyperparameters, Figures S4-S6). Despite the particularities of each reaction, we can observe a dominance of parameters related to reaction conditions, mainly temperature and catalyst composition. While temperature is naturally expected to play a crucial role on conversion (output variable), given the generality of the Arrhenius equation, it is interesting to notice how the software is able to predict how the presence of particular compounds affects the reacting system. For instance, while an experienced researcher may already understand that Ni is an established active metallic phase for steam reforming, a newcomer to this area would be able to build a similar conclusion per our program. Such information would be especially helpful for incipient researches as well as for predicting the result (in the case of steam reforming, methane conversion) for a new catalyst formulation and/or reaction condition without actually going through the steps of synthesizing the material and in fact conducting a bench-scale test. This approach could be useful particularly for high-throughput studies and for costly experiments, such as assays encompassing labeled reactants, precious metals, etc.

Table 3
Overall performance for each study applied for WGS reaction.

Study	R^2	Error
Cavalcanti et al. [29]	0.914	0.00909 (MSE)
Odabaşı et al. [17]	0.905	10.05 (RMSE)
CatS	0.935	0.0519 (MAE)

The reaction with the highest coefficient of determination was the steam reforming of methane with 0.941 and a MAE of 0.0464. The neural network for the WGS reaction obtained R^2 of 0.935 and MAE of 0.0519, metrics that surpassed those of previous studies, suggesting an adequate validation of the optimization process. Table 3 presents the

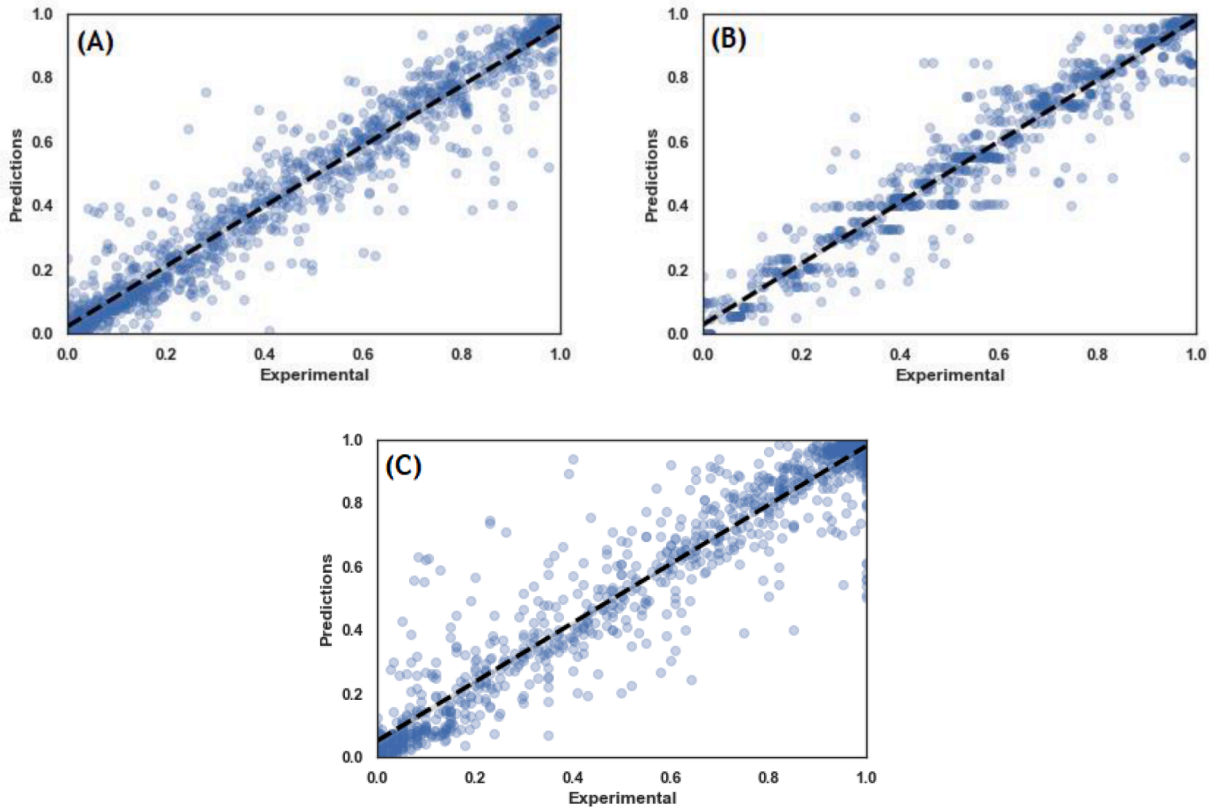

Fig. 6. Parity plot for WGS (A), Steam reforming of methane (B) and SelOx (C).

relation of coefficients of determination of each compared study. When compared to the model proposed by Cavalcanti et al. [29], the number of data points proved to be an important variable. When compared with works that used the same dataset (Odabaşı et al. [17]) the optimization process proposed a different architecture, which may affect the inferences result. It is paramount to highlight that we do not propose a final approach to the systems of interest and it comes to the user's finality. For instance, Cavalcanti et al. [29] conducted an exhaustive sensitivity analysis to their WGS analysis. While this certainly is of interest for those with this particular reaction in sight, it is outside of the scope of CatS, i.e., conducting general screening procedures. CatS' ability to spare time and resources is the main feature of our software, but this new catalyst screening method may be expanded to varied sensitivity analyses. Some paths that could follow CatS' results are: conclusions on the relation between reaction parameters; analysis of the importance of each parameter for the experiment; or analysis of the behavior of this reaction under adverse conditions The focus of this study is not to particularize any reaction, but to cover more reactions and areas within HC.

Two points should be highlighted about our findings. The first one is that the main factor that led the models to achieve admissible metrics was the successful combination of an expressively sized dataset with the optimization process described above. The large amount of reaction samples exposed the neural network to different situations, which made the learning more dynamic and prepared. In addition, the testing and validation of several network architectures brought the fine-tuning that the model needed to deal with eventual outliers and overfitting problems. From this finding and based on previous studies [30,31], we can conclude that the performance of a NN is directly related to the quantity and quality of data, especially when dealing with a model that infers results from chemical reactions.

The second crucial point about these metrics is that they are not static, that is, the dataset can continue to be populated and the model retrained. If a new technology is developed and outperforms the existing model, nothing prevents an adaptation in order to integrate the new feature to the application. Updates to existing models, as well as the implementation of other reactions, will be released according to community demand. On the other hand, the intrinsic randomness of experimental results, in addition to experimental problems, such as the standardization scarcity of experimental reports, difficulty in reproducing experimental results and human error in general are factors that can harm the network training and reduce its accuracy. An example germane to the HC community is the heterogeneity in which selectivities are reported. Some groups opt to express it as an analogous to mole fractions, while others present it regarding specific chemical elements of interest (for instance, carbon selectivity) and even as a ratio between the formation rate of a particular compounds over the sum of the formation rates over all components. Particular attention should be given to such potential discrepancies when feeding the software.

Fig. 6 presents the parity plot for each implemented reaction, where only the validation samples are shown, that is, 30% of the dataset. This plot complements the application output and shows the relationship between the values inferred by the model with its experimental references. Despite the high correspondence, it is still possible to identify some points that come out of the line on WGS and SelOx plots, which are perfectly normal in this context. These data points can represent experimentation error or unstudied conditions, that can be studied in future works and may provide promising catalyst combinations. An exaggerated fit in these graphs can represent a model overfitting, which occurs when a model specializes too much in the training dataset and when outside data is presented the model does not perform as expected. Such situations are usually avoided by a proper split between testing and training sets.

Another curious point are the horizontal lines that appear on the steam reforming plot, with some small occurrences in the initial portion and a dominant occurrence in the [0.2, 0.6] interval. These occurrences

could represent the existence of a reference catalyst and set of environmental conditions in the dataset, that is, a material that is widely employed throughout different studies under or comparatively between studies. For the particular case of steam reforming (Fig. 6B), the horizontally aligned data points represent studies with a benchmarked Ni/Al_2O_3 material. Once again, while this feature may be obvious to an experienced researcher, it could on the other hand be inconspicuous, for a newcomer, among a dataset containing thousands of reported results.

It is paramount to emphasize that the software is intended to assist operational research, and not to fully replace experimental steps. To validate CatS inferences, they need to be experimentally tested and reported back to CatS *a posteriori*, thus increasing the software's accuracy. Even with the advances of these technologies, it is necessary to pay attention to the fact that no model is able to replace the user critical sense. Experimentation will always overcome inference. Although the software has not been hitherto coupled to the experimental field, we can conclude from previous studies that machine learning techniques are already strong tools to complement these studies [32–34].

4. Concluding remarks and software outlook

We presented a way to apply ML techniques in the HC context in a simple and practical way, with scant coding knowledge required from the user. The results provided satisfactory fit between model prediction and experimental results (R^2 above 0.90) and indicated the most important variables for the assessed reactions, with temperature being relevant for all assessed reactions. Therefore, the findings could assist researchers in making decisions based upon previously reported results. In addition to the interface that compiles the models for each reaction, a generic procedure was developed to generate and optimize them.

Additionally to the approach of predicting the reaction result from a vector of inputs, our current research aims at determining catalyst and reaction conditions from a given output set point. We call this technique reverse engineering. Moreover, according to the acceptance from the scientific community, new models can be implemented.

Although the application has not been validated in experimental works, the implemented NNs presented satisfactory metrics that surpassed previous works. With CatS, the user will reduce the operational costs as well as reduce the time spent searching the best combination of catalysts. The software scope can be expanded to encompass online, accessible and community-powered databases, including studies that validate results obtained from computational sources (e.g., DFT, simulated particles or reactions) and not just empirical ones.

CRediT authorship contribution statement

Gabriel Sardinha de Oliveira: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization. **Henrique Poltronieri Pacheco:** Conceptualization, Methodology, Validation, Writing – review & editing, Supervision, Project administration.

Declaration of Competing Interest

The authors declare no conflict of interest.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.mcat.2022.112430.

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